A Seed Agent That Improves Its Own State: A Self-Contained Account for Readers With No Background in the Field

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Abstract

This paper describes a computer program that gets better at solving a kind of puzzle over time, and that keeps a change to itself only after that change has passed a strict test. The program is called a *seed agent*, because it is meant to be the thing that grows. Its rules for learning never change. What changes is what it carries from one round of experience to the next: a record of its own past guesses, a model of which puzzle ingredients tend to appear together, and the set of building blocks from which it assembles answers. Every proposed change is tried against the current version on puzzles that neither version has seen, and kept only when the improvement is large enough that luck is an unlikely explanation. The paper assumes no background in machine learning, statistics or programming. Every technical term is defined where it first appears and again in a glossary at the end. The results, in brief: on puzzle sets never used to decide anything, the main development agent's solving rate in its first puzzle setting rose from 17.1% to 28.8% over twelve rounds of experience, and it improved in four further settings with different hidden structure. A confirmation procedure that was fixed in advance, with four pass-or-fail conditions and a written forecast, was then run on nineteen fresh random seeds across ten versions. All 89 agent-and-setting pairs of those sealed runs improved, 169 of their 171 accepted changes were reproduced exactly by an independent replay (the other two had a wrongly recorded number, not a wrong decision), and three of the ten versions passed all four conditions on both of their seeds, at three, six and seven settings. A final pair of versions let the agent change its own learning rules for the first time, from a fixed menu of settings, keeping a change only when it passed a test of the same strict kind; the second of them was then run on ten fresh seeds. On all ten, on the one sample each was scored on, every setting and every group of puzzles improved, and every accepted change replayed exactly; nine of the ten passed the four conditions, and on the tenth one setting rose 5% where 15% was required. Nine of the ten also changed one of their own settings, 13 changes in all, of which one made the agent learn better and twelve only made learning cheaper. Finally, seven variants of the program were run on three of those seeds, under rules written down before any of them ran, to measure what each part is worth. Without its learned search guidance the program kept 6% of its growth, and without inventing building blocks 65%. The strict test added no growth: keeping every proposed change grew as much for less than a third of the computing, and spending the test's computing on more experience grew more. The paper therefore withdraws its earlier claim that the test is its central contribution. What the test did buy was safety: far fewer of the changes it kept made things worse. A second sample of puzzles, scored afterwards, confirmed each of these readings. The paper lists the limits that an audit of its own records found. The largest is that the agent starts without building blocks that every setting uses, so about a third of its measured growth is open to a start that was simply not held back. The paper also records what went wrong along the way, because each failure changed the design, and that record is as much the result as the growth is.

1 What this paper is about

Most programs that learn have two parts. One part is the *procedure*: the fixed rules that say how to turn experience into knowledge. The other part is the *state*: the knowledge itself, which the procedure reads and writes. A cook who follows the same method for learning new dishes, but whose notebook, pantry and sense of which ingredients go together all improve over the years, is improving state while keeping procedure fixed.

This paper studies a program of that kind. Its procedures for learning are fixed code that never changes while it runs. Its state is everything it carries between rounds of experience. The question is whether one program can improve its own state in several different puzzle settings, test every proposed improvement before keeping it, and carry what it keeps into settings with different hidden structure without damaging what it already learned.

The question is worth asking because carried improvements are not automatically safe. A change that helps in one setting can quietly hurt another, and a change that looks like an improvement on a small sample can be luck. Both of these happened in this work, and both led to changes in how the program tests itself. The paper is as much about the testing as about the growth, and Section 7.5 measures what the testing actually buys.

A word on terms before anything else. In this research programme, *recursive self-improvement* means a system producing a successor whose learning procedure is better than its own. That is not what happens here. Here the state improves and the procedure does not. The programme's own definition is therefore not met, and the paper says so plainly in Section 9.

The word *seed* has two meanings in what follows, and the paper keeps them apart. A *seed agent* is the program plus its carried state. A *random seed* is a number, such as 644101, that fixes the sequence of pseudorandom choices a run makes, so that the same run can be repeated exactly. Section 6 explains why that matters.

What came before. Three forms of growth had each been shown separately in this programme, in separate programs. First, a program that keeps the building blocks of whatever answer it submits, right or wrong, raised its own solving rate four to ten times across a hundred puzzles, and being told whether it was right added nothing. Second, a grouping of building blocks learned without any labels from 1,600 puzzle worlds let the same program solve 6,993 and 8,077 more puzzles over 18 budget settings than a grouping learned from 400 worlds, on random seeds fixed in advance, with the result independently checked. Third, in a small language for transforming text, a program that built new building blocks out of its failures solved 20 of 24 puzzles where its fixed vocabulary solved 5, on two random seeds. This paper puts the three forms into one program and asks whether they can be carried together.

2 The puzzles, from scratch

2.1 Characters, codes and rules

The puzzles are about characters. A *character* here is one of 256 symbols, each identified by a whole number from 0 to 255 called its *code*. The letter A, for instance, has code 65. A *rule* is a yes-or-no question about a character, such as "is its code below 128?" Every rule can be written as a table with 256 rows, one per character, saying yes or no.

Some rules are easiest to state using *bits*. A code written in binary is a string of eight digits, each 0 or 1, numbered from 0 on the right to 7 on the left. "Bit 7 is 0" and "the code is below 128" are the same rule, because bit 7 is 1 exactly when the code is 128 or more.

2.2 What one puzzle looks like

Each puzzle, called a *task*, hides one rule. The agent is shown a few example strings, and for each string it is shown how long the answer is: the answer is the longest run of characters from the start of the string that all satisfy the hidden rule. The agent's job is to find a rule that fits every example. It then submits that rule. A separate set of *hidden* examples, which the agent never sees, decides whether the submitted rule is actually right.

Here is one task from the implementation. The hidden rule is "bit 7 of the code is 0". The agent sees eight examples, of which the first three are:

answer prefix	rest of the string
119, 115, 48, 47, 121	152, 242, 205
23, 10, 101	168, 131, 144, 143, 137
61, 7, 118	240, 178, 187, 203

Every character in each answer prefix has a code below 128, and the first character after each prefix has a code of 128 or more. A candidate rule is *consistent* with an example when every character of the answer prefix satisfies it and the first character after the prefix does not.

2.3 The catalog and the budget

The agent does not invent rules from nothing. It holds a *catalog*: a long list of candidate rules built from its building blocks, which are described below. At the start of this work the catalog has 82,758 candidate rules. Of these, 77 are consistent with all eight visible examples of the task above, and 17 of those 77 also give the right answer on all sixteen hidden examples.

The agent cannot look at all 82,758 rules for each task. It has a *scan budget*, 1,024 rules per task throughout this paper, and it looks only at the first 1,024 rules in its current ordering of the catalog. The order is the thing the agent learns to improve. Looking at one candidate rule is called a *scan*.

In the task above, scanning the catalog in its starting order, the agent met exactly one consistent rule within its 1,024: "the code is below 144". It submitted that rule. The rule fails on a hidden example, so the task counts as unsolved. The agent is not told this. The right rule, "bit 7 is 0", was in the catalog all along, but outside the 1,024 rules inspected.

Because the agent found exactly one consistent rule in its window, this submission is what the paper calls a *confident pick*. A confident pick is not a guarantee. On average such picks are right about 90% of the time, and this one was wrong. Confident picks matter because the agent trusts them more when it learns from its own history.

2.4 Building blocks: atoms, families and the language

The smallest building block is an *atom*: one simple yes-or-no rule about a character, such as "bit 7 is 0" or "the code is below 144". There are 313 atoms in all, in 9 *families* named for the arithmetic operation they use. Examples of families are *bit* (a particular bit is 0 or 1), *threshold* (the code is below some number), *divmod* (the remainder on dividing the code by some number), *gcd* (the greatest common divisor of the code and some number), *popcount* (how many bits of the code are 1) and *xormod* (a combination of bit mixing and remainders).

The set of atoms the agent currently holds is called its *language*. This is not a natural language; it is a vocabulary of rules. The agent starts with 260 of the 313 atoms. Two whole families, popcount and xormod, are withheld. This is deliberate: it creates puzzles whose rules the agent cannot express at all until it invents the missing atoms.

The catalog is built from the language. It contains every atom, the opposite of every atom (its *complement*: "bit 7 is 1" is the complement of "bit 7 is 0"), and every pair of atoms from different families joined by *and* (both must hold), *or* (at least one must hold) and *xor* (exactly one must hold). Rules that behave identically on all 256 characters are merged into one row. From the starting 260 atoms this gives 82,758 rows; from all 313 atoms it would give 166,596. Whenever the language changes, the catalog is rebuilt.

2.5 Two different ways to fail

The agent can fail a task for two different reasons, and the distinction organises the whole design.

1. The right rule is in the catalog but outside the budgeted window. The worked example above is this kind of failure. The remedy is a better ordering of the catalog, so that the right rules come earlier. The paper calls this *search guidance*.
2. The right rule cannot be built from the current language at all. A task whose hidden rule uses a popcount atom is this kind of failure for the starting agent. The remedy is to add the missing atom. The paper calls this *invention* or *extending the language*. In some puzzle worlds a third of the rules cannot be expressed by the starting language.

Two cautions. A task on which the agent finds no consistent rule does not prove that an atom is missing; it means only that the budgeted search found none. And invention here is not unrestricted creativity: the agent assembles candidate atoms from a small fixed grammar of arithmetic expressions, described in Section 5.

3 How the puzzles are organised: worlds, areas, pools and lives

Worlds. Tasks do not come one at a time from nowhere. A *world* is a hidden set of 14 atoms, and every task in that world has a rule built only from those 14. A world has 100 tasks. The tasks of a world are therefore related, as dishes from the same pantry are related, and an agent that notices which atoms a world uses can order its catalog to favour rules built from them. Think of a world as a cuisine, its 14 atoms as its pantry, and each task as a dish.

Areas. An *area* is a recipe for generating worlds, with a particular hidden structure deciding which atoms go together. Four kinds of area are used. Their names in the code are given in parentheses.

- **Mechanism worlds** (`mech`). A world draws three of seven atom families and takes its 14 atoms from those three. The agent's starting language lacks the two remaining families, so a world that uses one of them contains rules the agent cannot express. A fifth of the possible family triples are withheld from the agent's experience as *untrained combinations*; they appear only in evaluation. The agent's experience draws three quarters of its worlds from trained triples and a quarter from triples that include a family the language lacks.
- **Drawn partitions** (`drawn7`, `drawn11`, `drawn13`). Every atom of every family is assigned at random to one of a few groups, with no regard for family. A world picks some groups and draws its 14 atoms from them, and each atom has some chance to *slip* to a random atom instead. Because the grouping cuts across families, nothing the agent learned about mechanism worlds helps here except its language. The three areas differ in structure: 9 groups with 3 per world and a slip chance of 0.15; 12 groups with 4 per world and no slip; 6 groups with 2 per world and slip 0.15. The agent is never told how many groups a world uses.

- **Metric worlds** (`metric777001`, `metric555001`). Each atom is given a numerical fingerprint describing how its yes-or-no answers are distributed across the 256 characters. A hidden weighting over the parts of that fingerprint is drawn once and shown to nobody. A world is one centre atom together with the 13 atoms whose fingerprints are nearest to it under the hidden weighting. Here the hidden structure is about how atoms behave, not how they are written. A quarter of each round's worlds are centred on an atom of the two withheld families. The two areas use two different hidden weightings.
- **String tasks** (`str7`). A different kind of puzzle entirely, added late in the work to test whether the agent's machinery depends on the character puzzles. A task is three visible and six hidden pairs of an input string and an output string, produced by a hidden small program. A program glues together up to three pieces; each piece is either the part of the input between two positions, optionally changed to upper or lower case, or a fixed constant. The positions, the three case operations and six constants are the building blocks here, called *keys*: 58 in all, from which 9,400 candidate pieces are built. A world is 14 keys from three of seven key families, with 100 tasks over programs that use only those keys. The twelve positions that refer to runs of upper-case or lower-case letters are withheld from the starting language, which therefore has 46 keys and 5,334 pieces. A task is solved when the first program in the agent's budgeted window that reproduces the three visible pairs also reproduces the six hidden ones.

Pools. Within an area, worlds fall into *pools* that the scoring can tell apart but the agent cannot. Mechanism worlds and string tasks have three pools: *trained* (family combinations the agent lives in), *untrained* (combinations withheld from its experience) and *unfamiliar* (a family its starting language lacks). The drawn and metric areas have one pool each. Pools let the evaluation ask whether the agent got better only where it practised, or everywhere.

Lives. A *life* is one round of experience followed by one round of self-change. In the development runs a life is 48 worlds of one area, 100 tasks each; in the confirmation runs a life is 96 worlds. Only after living all of them does the agent propose changes to itself and test them. Lives in different areas can be interleaved, because the agent saves its state after every step.

What the agent is never told. It is never told which family an atom belongs to, what a world's true 14 atoms are, which pool a world came from, or whether any particular submission was right. Those labels are used only by the scoring, after the fact.

4 What the agent carries, and what stays fixed

The *carried state* is everything the agent keeps from one life to the next:

- its **language**, the atoms it currently holds, including any it invented;
- its **experience**, one record per world it has lived, naming the area and the atoms behind its confident picks and behind all its picks, stored by atom name so that the records survive changes to the language;
- for each area, a **world and self model** and a flag saying whether to build a **search prior** from that area's experience, both explained in the next section;
- the list of **invented atoms** and the life each entered;

- a **ledger** recording every proposed change, the test it faced and its outcome;
- a **resource account** of everything spent on living, testing and inventing.

The language, the experience and the ledger are shared across all areas. The model and the prior belong to one area each. This sharing is where the danger lies: a language change made for one area changes the catalog that every other area's model was fitted over.

The fixed part is the code of the five operations in the next section. Through version 10, nothing the agent does alters how that code behaves. When the acceptance rule changed during this work, the experimenter changed it, and the paper records each change. From version 11 the agent also carries its **procedure**: six numbers that tune its own operations, such as how many new atoms it may invent in one life. The code still never changes, but the agent can change those numbers, under a test, and Section 7.3 describes how.

5 The five operations

Live: collect evidence while solving related tasks. The agent draws fresh worlds of an area and works through each as a stream of 100 tasks at the scan budget. Within a world, the atoms of each answer it submits are moved towards the front of its catalog order for the next task, so that a world's own pantry is discovered progressively. Two carried objects reorder the catalog further. The area's model, when there is one, turns the atoms earned so far in the current world into beliefs about which groups of atoms this world uses, and promotes the rows those groups build. The area's prior, when there is one, starts the scan from an order based on how often atoms appeared together in the agent's own earlier worlds of that area, with half of each task's budget reserved for the order the agent would use with no prior at all. With neither, the agent scans in catalog order. It receives no feedback on whether any answer was right.

Reflect: turn accumulated evidence into better search guidance. After a life, the agent looks back over every world it has lived in that area and fits a *world and self model*. The "world" half asks which atoms tend to occur together; the "self" half asks how reliably the agent finds each atom when it is present, and how often it mistakenly picks up an atom that is absent. In detail, the model supposes that atoms belong to hidden groups and that each world uses a few groups. It fits, from the agent's confident picks alone, which group each atom belongs to, how often each combination of groups occurs, and for each atom two rates: a *discovery rate*, how often the agent finds the atom when a world contains it, and an *import rate*, how often the agent earns the atom through a wrong pick in a world that does not contain it. The fitting uses a standard iterative method called expectation-maximisation, which alternates between guessing the hidden group labels and re-estimating the rates until the two agree. An atom whose discovery rate falls below 0.10 is filed in no group: the agent meets it only through its own mistakes, and that verdict is used later to clean up the language. The model's size, 6 to 8 groups and 2 to 4 groups per world, is chosen by how well each candidate size predicts the most recent fifth of the agent's worlds in that area. The refit model, together with a prior built from the area's experience, forms one proposed change.

Extend: propose new atoms from the tasks the agent could not cover. For the third of the life's worlds with the most tasks on which no consistent rule was found, and only those worlds with at least 15 such tasks, the agent enumerates small arithmetic expressions from a fixed grammar, 3,456 candidates distinct up to complement, searched smallest first. Each candidate is tested against the uncovered tasks, alone, complemented, or joined by and, or or xor to one atom the agent earned

in that world. A candidate is proposed only if it fits at least two uncovered tasks. The reason is recurrence: an atom a world is actually built from shows up in several of its rules, while a chance fit shows up once. At most 40 atoms are proposed per life. The first version of this operation accepted single fits, and every extension it proposed lost its test.

Prune: propose removing inventions that never paid. An invented atom is proposed for removal two or more lives after it entered if it has never appeared in a confident pick, or if every area model that has seen it rates it as found only through wrong picks.

Prove: decide whether a change deserves to persist. This is the operation the whole design turns on. Its name is a misnomer the implementation kept: it is an empirical test, not a mathematical proof, and the paper otherwise says *test* and *accept*. After a life, the agent proposes all three kinds of change at once and tests the combination. If the combination fails, it tests each part on its own. A test works as follows.

1. A *held-out set* of worlds of the area is drawn for that life. "Held out" means the agent never lived these worlds and they are used only for comparison. In the development runs the set has 24 worlds per pool; in the confirmation runs it has at least 48.

2. Both the current state and the candidate state are run on every world of the set, and for each world the difference in tasks solved is recorded. Because the two states are compared on the same worlds, this is a *paired comparison*: world-to-world variation in difficulty cancels out.

3. The per-world differences are summarised by a *95% bootstrap interval*. The procedure resamples the list of differences with replacement 4,000 times, computes the average of each resample, and reports the range that contains the middle 95% of those averages. If that whole range lies above zero, the candidate's gain is unlikely to be luck, and it is accepted. If the range includes zero, the data cannot tell a real gain from noise, and the candidate is rejected. A candidate that is clearly behind after the first half of the set is stopped there, to save cost.

4. A candidate that changes the language is also run on a smaller held-out set of every other area the agent has lived in, 12 worlds per pool in development and 24 in confirmation, and is rejected if its total there falls more than 2% below the current state's. This is the *cross-area check*, added after the failure described in Section 8.1. A language change is checked only against areas of the same puzzle form, since the character and string forms build their catalogs from disjoint building blocks.

5. From version 10 of the confirmation protocol, a candidate whose interval includes zero but whose average gain is large enough that doubling the sample would be expected to clear zero is run once more on an *extension set*, a second held-out set of the same size drawn fresh, and the decision is made on the two sets together. This was added after the failure described in Section 8.3.

6. Every world run in any test is charged to the agent's resource account, so that the cost of testing is visible and can be gated.

The agent therefore receives feedback at exactly one point: aggregate scores when it decides which changes to keep. It never receives feedback on individual answers.

Figure 1 shows the cycle. The boxes are the fixed procedures. The state on the return path is what changes.

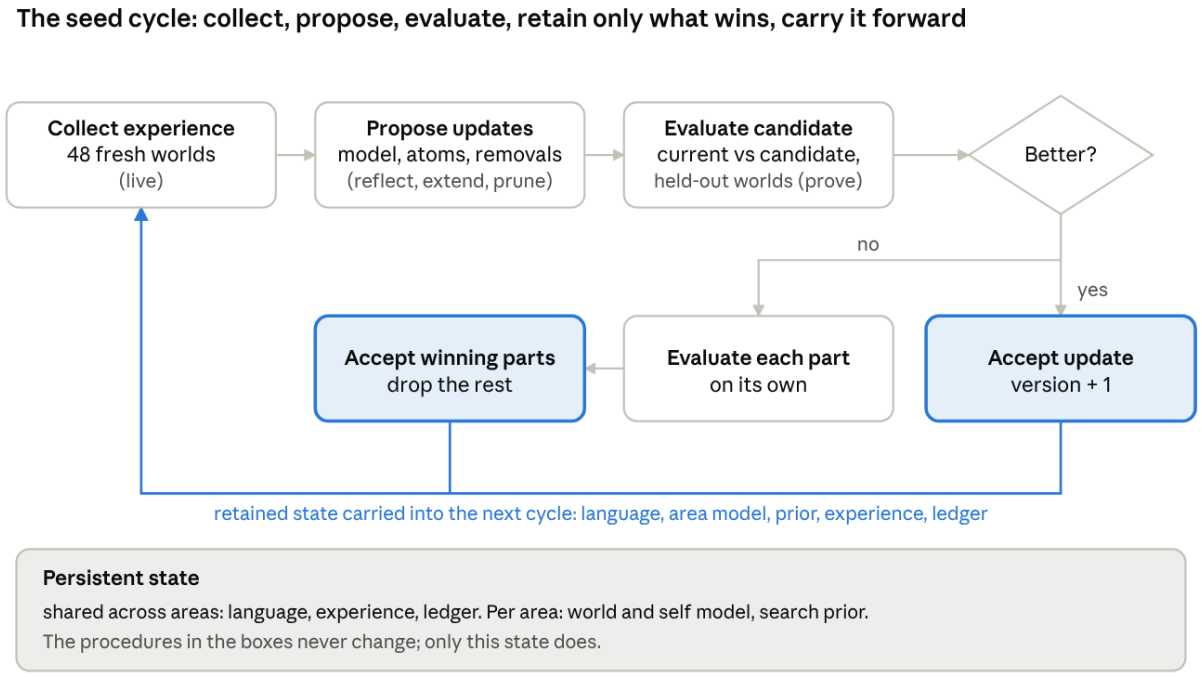


Figure 1: One life of the seed agent. It collects experience in fresh worlds, proposes changes of three kinds, tests the combined candidate against the current state on held-out worlds drawn for that life, accepts it if it wins and otherwise tests each part alone, and carries the retained state into the next life. The language, experience and ledger are shared across areas; the model and prior belong to an area. Implementation names in parentheses.

6 How the result was made trustworthy

A program that improves on the puzzles it was designed against proves little, because the designer may have tuned it to those puzzles. The work was therefore split into two phases with different rules.

Development. Three agents were run on random seeds 644101, 643101 and 645101, named `s1`, `s2` and `s3`. Their results were inspected as the design changed, and the acceptance rule was revised four times in response to what they showed. These results are informative but not evidence in the strict sense, because the design was adjusted while looking at them.

Confirmation. Once the design was settled, a protocol was written down and frozen. Freezing means four things.

- **The code was hashed.** A *hash* is a short fingerprint computed from a file's exact contents; changing a single character changes the fingerprint. The fingerprints of the source files were recorded, so that anyone can check that the code that ran is the code that was frozen.
- **The random seeds were sealed.** A *random seed* is the number that determines every pseudorandom choice a run makes: which worlds are drawn, which held-out sets, which resamples.

Running the same code with the same seed reproduces the same run exactly. Sealed seeds, 919100 onward, were chosen in advance and never used before, so the design could not have been tuned to them.

- **A forecast was committed.** Before each version ran, the probability that all four gates would pass on both of its seeds was written down. This keeps the experimenter honest about what was expected.
- **Four gates were stated in advance.** A *gate* is a pass-or-fail condition. The four are: (1) *growth*, every area's score on a fresh held-out set rises by at least 15%; (2) *safety*, no pool of any area ends below 90% of its starting score; (3) *cost*, the worlds spent on testing are at most a stated multiple of the worlds spent living, 2.5 times in the three-area protocol and more when more areas are checked; (4) *audit*, every accepted change is reproduced exactly by an independent replay from the saved state, and no two acceptances in an area used the same held-out set.

The final check. The gates are scored on *fresh* held-out sets: one further set per area, 24 worlds per pool, drawn with a random salt no test ever used. A *salt* is an extra number mixed into the random draw so that a set is distinct from every other. On these sets the starting state (starting language, no model, no prior) and the retained state are run once each, and the two totals are compared. No acceptance decision ever saw these sets.

The audit. Because every run is determined by its seed, an *auditor* program can rebuild the state the agent had before and after each accepted change, draw the same held-out set, run both states again, and check that the totals and the interval come out identical to what the ledger recorded. If they do, the acceptance "replays exactly". This catches bookkeeping errors, and it caught two during the work.

Each version runs on two new seeds. Every change to the protocol, however small, was frozen as a new version with new hashes and run on two seeds never used before. A version passes only if all four gates pass on both. Table 1 summarises which information each stage of the system is allowed to use.

Table 1: What information is used where. The final evaluation sets were never used to accept anything.

Stage	Information used	Decision it supports
Solving tasks	Visible examples, own submissions, how many consistent rules were in the window	Search order and experience collection; no feedback on correctness
Choosing the model's size	How well each size predicts the most recent fifth of the area's lived worlds	Number of groups
Accepting a change	Aggregate held-out scores, paired over worlds	Whether a change is kept
Final evaluation	Further held-out sets never used for acceptance	The reported gains and the gates

Table 2: Development agents: tasks solved on held-out sets never used by an acceptance test, 24 worlds per pool, scan budget 1,024.

Agent (random seed)	Area (lives)	Start	Retained	Tasks
s1 (644101)	mechanism worlds (12)	1,232	2,072	7,200
s1	drawn7, 9 groups of 3 (4)	211	364	2,400
s1	metric worlds (4)	427	2,076	4,800
s1	drawn11, 12 groups of 4 (3)	199	360	2,400
s1	drawn13, 6 groups of 2 (2)	294	442	2,400
s2 (643101)	mechanism worlds (3)	1,292	1,879	7,200
s2	drawn7 (2)	221	424	2,400
s2	metric worlds (2)	492	1,520	4,800
s3 (645101)	mechanism worlds (3)	1,081	2,074	7,200
s3	drawn7 (2)	280	246	2,400
s3	metric worlds (2)	394	1,349	4,800

7 Results

7.1 Development: did the three agents improve?

In ten of eleven agent-and-area pairs, yes. Table 2 gives the counts on held-out sets never used for acceptance. The main agent `s1` lived twelve lives in mechanism worlds, four in `drawn7`, four in metric worlds, three in `drawn11` and two in `drawn13`. Its mechanism-world solving rate rose from 17.1% to 28.8% (1,232 to 2,072 of 7,200 tasks), with one qualification: trained family combinations rose from 26 to 49 tasks per world and worlds with an unfamiliar family from 5 to 19, but untrained combinations fell from 20.5 to 18.6. Section 8.2 explains why. Agents `s2` and `s3` lived three mechanism lives and two lives each in `drawn7` and metric worlds.

The one exception is `s3` in the drawn partition, where the retained state scored 246 against a start of 280. The loss came from a 40-atom language extension accepted in metric worlds, which passed its cross-area check on the drawn partition on a 12-world set (124 to 148) and then lost on the larger 24-world evaluation set. The lesson, taken into the confirmation protocol, is that a 12-world check is too small.

Which kinds of change were kept. Refit models were the most consistently accepted: on `s1` a refit model was accepted on every mechanism life from 0 to 6 and on the first tested life of every other area, and on `s2` and `s3` on all three mechanism lives. Language extensions were accepted less often, 6 of 24 proposed on `s1`, and every extension proposed on `s2` and `s3` in mechanism worlds was rejected. Four removals were accepted on `s1`, taking out 23, 26, 25 and 55 invented atoms. Over its 26 versions, `s1` accepted 16 models, 6 extensions and 4 removals, and rejected 18 extensions, 7 models and 1 removal. These are counts of decisions, not a breakdown of where the gain came from; Section 7.5 gives that breakdown.

What was invented. At its 26th version, `s1`'s language held 111 invented atoms. Of these, 26 reproduce true popcount atoms and 29 true xormod atoms, the two withheld families, and 56 behave like nothing in the original 313. Every invented atom entered through an accepted test. In the drawn areas the agent's model chose four groups per world for the 12-group area, which is the truth, and four for the 6-group area, where the truth is two.

What it cost. By version 26, `s1` had spent 122.9 million scans living, 302.3 million scans testing, and 831,738 invention tests. Testing cost two and a half times living, because each life tested up to four candidates on 72 worlds against a freshly scored current state, with cross-area checks on top.

7.2 Confirmation on sealed seeds, version by version

The protocol went through twelve versions. Each was frozen with new hashes and run on two new sealed seeds, and seven of the eleven revisions came after a gate failed; the other four (versions 5, 8, 9 and 11) extended or repaired the design after a pass. Table 3 summarises the sequence; Tables 4 and 5 give the scores; Figure 2 plots every agent-and-area pair.

Version 1: the test was too expensive. The first frozen protocol passed growth, safety and audit on both seeds and failed cost on both, at 5.96 and 5.88 times living. The arithmetic: drawing a fresh held-out set every life means re-scoring the current state every life, which alone costs 1.5 times a 48-world life in the mechanism area, and three candidates per life come on top.

Version 2: cheaper tests, and a bookkeeping flaw. The schedule's economics changed and the acceptance rule did not: lives of 96 worlds, held-out sets of 48 worlds in every area, a held-out set kept until something is accepted on it, and a stop at the half set when the average gain is not positive. Cost fell to 1.89 and 1.98. But on one seed two parts of one life were accepted on the same held-out set, which the audit clause forbids, because an acceptance on a set makes that set no longer fresh for the next candidate.

Version 3: a stale cache. A fresh set is now drawn after any acceptance within a life. The audit then found one acceptance per seed whose recorded baseline, the current state's score, was off by 47 and 27 tasks. The cause: the current state's score on a reused set had been cached, and as the area's experience grew, the prior built from that experience changed, so the cached score went stale. Every acceptance still stood when replayed with the correct baseline; the recorded number was wrong, not the decision.

Version 4: confirmed for three settings. The cache is now keyed on the experience count as well as the state version. All four gates passed on both seeds, at 2.32 and 2.04 times living. Across the eight seeds of versions 1 to 4, every area rose by at least 17% and every pool rose except one, which ended at 94% of its start. The cross-area check rejected shared changes on most seeds for losses on the mechanism area that the proposing area's own test had accepted.

Version 5: a guard that did not work. The development runs had shown the shared language diluting the search on untrained combinations (Section 8.2). Version 5 added a guard: a candidate that adds atoms is also scored with the model and prior switched off, and rejected if that unguided score falls more than 10% below the current language's. On seed 919108 the guard never fired, three extensions passed at unguided losses of 2% to 9% each, and the untrained pool ended at 89% of its start while the drawn partition grew only 6%: growth and safety both failed. On seed 919109 the guard fired three times, every pool rose, and testing cost 2.53 times living, over the gate. A per-candidate tolerance does not bound what several candidates do together, and the guard was retired. Version 4 remained the confirmed protocol for three settings.

Versions 6 and 7: six settings. Both kept version 4's code byte for byte and widened the schedule to six areas, adding two drawn partitions and a second metric weighting that no agent had lived before. Version 6 passed growth, safety and audit on both seeds, with every one of the twelve area scores and twenty pools rising, and failed cost on both at 2.78 and 2.64: a language candidate that passes its own area is then checked on every other lived area, so the test bill grows with the number of areas, by about 0.18 per area over version 4's mean. Version 7 changed only the cost threshold, to 2.5 plus 0.25 per area beyond three, the 0.25 being the structural ceiling of one 24-world check against a 96-world life per candidate. It passed all four gates on both seeds, at 2.88 and 2.32.

Version 8: a second kind of puzzle. The string-task area was added, with three lives, 21 lives in all and the cost gate at 3.5. All four gates passed on both seeds: fourteen of fourteen area scores rose, by 27% to 356% with a median of 78%, 26 of 26 pools rose, 24 of 24 acceptances replayed, and testing cost 2.61 and 2.71 times living. The string area rose 27% and 32%, and both seeds invented the same six withheld positions, the end of the first, second and third upper-case run and the start of the first, second and third lower-case run, proposed from what stuck worlds had earned and accepted on held-out tasks with no label read.

Version 9: a repaired filter, and a miss at the noise floor. The string worker's test for whether a candidate piece is consistent with the visible examples had excluded any piece that returns an empty string on some visible input. The puzzle generator's own pieces do exactly that when two positions coincide, so 283 of 2,400 development tasks had no expressible program at any budget. With a piece allowed to be empty on some inputs but not all, 16 such tasks remained, and the same carried state solved 2,015 of those tasks instead of 1,901. On the sealed seeds the string area started higher, at 5,076 and 4,910, and still rose 23% and 25%, with eleven of the twelve withheld positions invented on each. But seed 919116 failed the growth gate in `drawn7`, at +8%, an area the change did not touch. Section 8.3 examines that miss. Version 9 fails as a protocol; the repair itself held.

Version 10: a second look at borderline candidates. A candidate whose interval includes zero, but whose average gain is large enough that doubling the sample would be expected to clear zero, is now run on one extension set of the same size and judged on both sets together, and the auditor replays both. The first sealed seed passed all four gates at 2.92 times living; two of its fifteen acceptances were made on an extension set, and both were confirmed there at a gain similar to the first set's. The second seed was interrupted by the host machine, which stopped the run for lack of memory at the twentieth of 21 lives. That seed was never rerun and left no data; a resumed run would have skipped one life's tests and would have to be recorded as a deviation.

Forecasts. The probability written down before each version ran, that all four gates would pass on both seeds, was 0.15, 0.35, 0.4, 0.3, 0.15, 0.2, 0.35, 0.3, 0.35 and 0.3.

7.3 Versions 11 and 11.1: the agent changes its own rules

Up to version 10 the agent improved what it knew but not how it learned. Version 11 gives it a small set of dials on its own learning rules and lets it turn them, one click at a time, under a test. There are six dials. *Patience* and *probation* control how quickly the agent stops trying a kind of change that keeps failing, and for how long. The *extension cap* is the most new atoms it may invent in one life. *Recurrence* is how many stuck puzzles a new atom must fit before it is proposed. *Share*

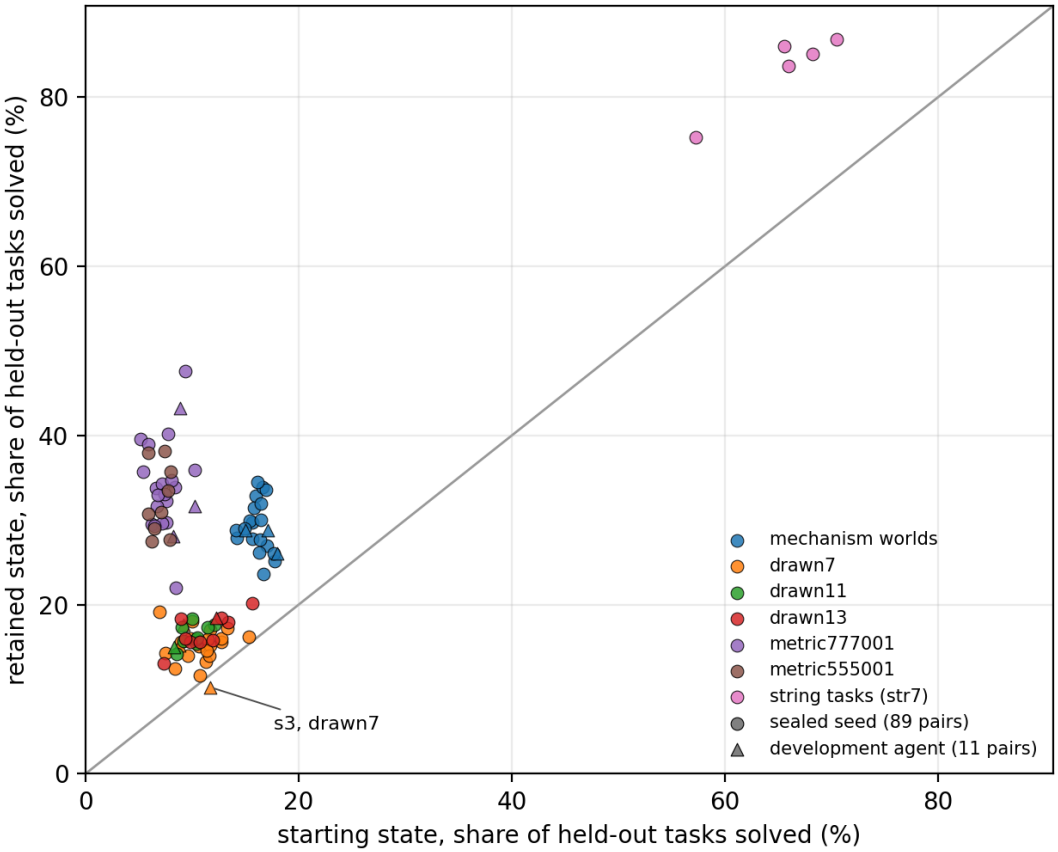


Figure 2: Share of held-out tasks solved on sets no acceptance test used: the starting state (horizontal) against the retained state (vertical), one point per agent and area, 100 pairs in all (the three development agents in eleven, the nineteen sealed seeds in 89). A point above the diagonal improved. One pair fell, `s3` in the drawn partition.

is what fraction of the hardest worlds invention looks at. The *pruning threshold* is how rarely the agent must find an atom before it proposes removing it. Each dial has two or three settings.

Every second life, the agent reads its own ledger to decide which dial to try and which way. For example, if it kept an invention that hit the cap, that is a vote for a bigger cap. Then it splits into two copies, called *branches*, that start from exactly the same place and live the same worlds: one under the current dials, one with the dial turned. After one life (two for patience and probation, whose effect only shows across lives), both copies are scored on a *meta set*, 48 fresh worlds that no other test has used. The turned dial is kept, along with that copy's state, if it solved more with an interval above zero, or if it solved no fewer and cost at least a tenth less to test. Otherwise the other copy carries on. These tests are paid for like everything else, and capped at 30% of all testing.

Version 11 ran on seed 919120. It passed growth, safety and cost, and it adopted two dial changes, each giving the same result for 15% to 17% less testing. It also turned down a third that would have saved 42% but solved fewer puzzles (683 against 730 on its meta set). Yet it failed the replay condition on that refusal. Over a two-life test, the losing copy's second life had produced experience that was never saved, so the independent replay could not rebuild it, and scored it at 699. Re-running that one life by hand reproduced 683 exactly: the agent's decision was right; its record was incomplete. Version 11.1 saves that experience and changes nothing the agent decides.

Version 11.1 passed all five conditions on both of its fresh seeds, 919121 and 919122 (the last three rows of Table 5). Every area rose, by 29% to 631%. Every pool rose. Testing cost 2.1 to 3.7 times living, inside the limits. Every accepted change, every rule revision and every dial test replayed exactly. Of eight dial changes tested, four were adopted: a bigger extension cap on both seeds and a lower pruning threshold on one, each the same result for about 15% less, and on 919122 a stricter recurrence that produced a better state, 759 puzzles against 724 on its meta set. That last one is the first time in this work that the agent changed a rule of its own and learned better for it.

7.4 The final version on ten seeds

A reviewer of an earlier draft pointed out a real weakness: the sealed seeds were spread over ten versions, and only one finished run used the final one. So version 11.1 was run, with its files unchanged, on eight more fresh seeds, 919127 to 919134. With 919121 and 919122 that makes ten seeds under one frozen version.

Every area and every pool rose on every seed (Table 6, where the *median* is the middle value of the ten and the *interval* says how far that middle value could plausibly move if the seeds had come out differently). Nine of the ten seeds passed the first four conditions (Table 7). Seed 919127 did not: one area, `drawn13`, went from 328 to 343, a rise of 5% where 15% was required. The written forecast had given only a one-in-five chance that all eight new seeds would pass, and seven did. The two pools named in advance as the ones to watch, the untrained pools of the first puzzle setting and of the string puzzles, rose on every seed, by at least 13% and 8%. The 9% fall on untrained combinations that the reviewer singled out is not from these runs. It belongs to the main development agent, 20.5 to 18.6 tasks per world (Section 8.2), and was read during development, before the four conditions were fixed and before any sealed seed was opened; it would have sat inside the 10% the safety condition allows. Testing cost 2.1 to 3.0 times living for changes to the agent's knowledge and 2.6 to 4.2 times once the dial tests are counted, inside the limits on every seed. All 123 accepted changes replayed exactly.

Nine of the ten seeds kept at least one dial change, 13 in all. Seven of the 13 were the same change, a bigger extension cap. Twelve gave the same result for less testing. One, a stricter recurrence on seed 919122, made the agent learn better. That is a much narrower result than the growth in the agent's knowledge, and Section 9 says so.

How big does the test need to be? The ledgers record the result of every world in 902 complete tests, so those tests can be judged again on only part of their own worlds, with no new computing (Table 8). On half the worlds the test would lose 29% of the changes that cleared the full test's interval, and pass only 6% of the ones that did not; on a quarter of the worlds it would lose 48% and pass 8%. So a smaller test mostly throws away good changes and rarely lets in ones the larger test refused. This compares a small test with the full test, not with the truth, because a change that was never kept has no later score to check against. The tolerance in the check on other areas matters a great deal: of 238 candidates that reached that check, 49 would pass with no tolerance at all, 80 at the 2% actually used, 152 at 5% and 211 at 10%.

7.5 What each part is worth

A reviewer asked two questions the results above cannot answer. How much of the growth does each part of the agent produce? And is the strict test worth its cost, which is two to three times the cost of living? To answer them, seven variants of the agent, called *arms*, each lived the full version 11.1

schedule on three of the ten seeds, 919127 to 919129, with the dial tests switched off. On each seed every arm started from the same state.

- **Full**: the agent as described.
- **Accept-all**: every proposed change is kept, with no test at all.
- **Point**: a change is kept if it solved more puzzles on half the held-out set, with no interval, no second sample and no check on other areas.
- **Matched**: every change is kept, and the computing that full spends on testing is spent on more experience instead, so that both use the same total.
- **No-guidance**: no model and no search prior are ever built.
- **Frequency**: no model; the search prior is just how often the agent has earned each atom.
- **No-invention**: no atoms are ever invented or removed.

Each arm's growth is summarised by one number, G: for each of the seven areas, the logarithm of the retained score divided by the starting score on the fresh held-out set, averaged over the areas. A G of 0.695 means the scores rose about twofold on that average. Before any arm ran, four rules for reading the outcome and a forecast were committed to the repository. Three seeds are too few for an interval, so the rules count on how many seeds full came out ahead. Table 9 gives the outcome.

Guidance and invention produce the growth. Full beat no-guidance, frequency and no-invention on all three seeds. Without any model or prior the agent reached a G of 0.045, 6% of full's, and kept one change in three runs. With only atom frequencies as its prior it reached 0.513; most of that gap is in the two metric areas, and in the other five full was ahead by little. Without invention the agent reached 65% of full's growth, and in the puzzles built partly from families the starting language lacks it solved about half as many in mechanism worlds (7 to 12 per world against 17 to 23) and about 30% fewer in the string puzzles (59 to 62 against 82 to 87).

The test buys safety, not growth. Accept-all and point each came out ahead of full on one seed of three, which the rules call inconclusive. Matched came out ahead on all three. The rule written in advance for that case says the paper must withdraw the claim that the test is its central contribution and must say that the test did not earn its cost at this budget and in these areas. This paper does both. Accept-all grew as much as full for 29% of the computing. What the test did buy is fewer bad changes. Counting a kept change as harmful when it lowered its own area's score on a fresh set, full kept 2 harmful changes out of 35, accept-all 15 out of 63 and point 17 out of 57. The test is not a guarantee either: the one change the no-guidance arm kept, 35 invented atoms, passed its test and the check on other areas and then cost the mechanism area 13.5% and `drawn7` 19% on their fresh sets. The forecast written with the rules scored 0.126 on the Brier scale, where always saying "50%" scores 0.25 and lower is better.

A second sample. All of the above rests on one sample of puzzles per area. The 21 retained states were therefore scored again on a second sample that nothing had used, with the scoring script committed before it ran (Table 10). Because this time every world's result was saved, each difference from the full agent carries an interval, and a star in the table marks a difference that luck is unlikely to explain. Every reading held: matched was ahead on all three seeds, accept-all and

point were level with full, and the three arms with a part removed were behind. Section 9.1 says what else the second sample showed.

7.6 The totals across nineteen sealed seeds

Every one of the 89 agent-and-area pairs of the sealed seeds improved on its fresh held-out set. 169 of the 171 accepted changes replayed exactly; the other two were the stale baselines of version 3, wrong recorded numbers rather than wrong decisions, and nothing else was found. Growth of at least 15% in every area held on seventeen seeds of nineteen; the two misses are seed 919108 under the retired guard and seed 919116's drawn partition at +8%. No pool below 90% of its start held on eighteen of nineteen; the one miss is 919108. Across the nine seeds that lived six or more areas, 59 of 59 area scores rose, by 8% to 562%, and 105 of 105 pools rose. The mechanism area's untrained pool, the one the development runs had flagged as the weak point, rose on seventeen of nineteen seeds to a median of 163% of its start; its two dips, to 94% on 919103 and 89% on 919108, are both under early versions, and its outcome shows no relation to how many shared-language changes the other areas kept.

8 What went wrong, and what each failure taught

Each of the following was a real failure in the record. Each changed the design, and the design is best understood through them.

8.1 A change in one area damaged another

After four lives in metric worlds, the main agent's score on the first drawn partition's untouched held-out set had fallen from 376 to 338, a 10% loss in an area it had not entered since. The cause was the shared language. The metric lives had accepted one extension of 40 atoms and one removal of 25, each tested on the metric area alone. A language that grew and shrank to suit metric worlds changed the catalog that the drawn partition's model had been fitted over. Each change was a tested success where it was tested and a regression where it was not.

The fix was the cross-area check of Section 5: a language candidate is also run on a held-out set of every other area the agent has lived in and rejected if it falls more than 2% behind there. The tolerance permits a small measured regression; it is a bound, not a guarantee of preservation. The check fired twice as soon as it existed. In `drawn11` a combination of invented atoms and a model gained 165 to 344 on that area's own held-out set and was rejected because it cost mechanism worlds 8% and `drawn7` 12%; the model alone was then accepted, at 417. In `drawn13` a combination worth 150 was rejected for costing `drawn11` 23%. Each would have been accepted under the earlier rule. The `s3` case of Table 2 shows the check's remaining gap: a 12-world set read a gain where a 24-world set read a 12% loss, which is why the confirmation protocol uses 24.

8.2 The grown language diluted the search

At version 24 the main agent's untrained family combinations had fallen from 24 to 18 tasks per world on the mechanism area's untouched set, while trained worlds rose from 26 to 50. The acceptance tests had not noticed, because a test compares totals and the untrained pool is only a third of the set.

Table 11 takes the retained state apart to find the cause. An *ablation* removes one component at a time and measures what is lost. With the retained model and prior but the starting language,

the state scores 1,973, with untrained combinations at 26 per world, above the start. With the grown language alone, no model and no prior, it scores 735 against 1,299 for the starting language. The 166 invented atoms then in the language, 110 of them behaviourally new, had enlarged the catalog from 82,758 rows to 266,272, and the half of each task's budget that the prior rule reserves for the no-prior order was scanning a catalog two thirds of which the agent had added. Trained worlds did not suffer, because the model's half of the budget found their rows anyway. Untrained combinations, which the model promotes less, did.

The agent already held the verdict it needed. Its self model files an atom in no group when the agent would almost never find it in a world. Across the five area models, 107 invented atoms were rated real by at least one model, 55 of the 56 that reproduce a withheld family among them, and 59 were rated import-only by every model that had seen them, one of them real. From version 26 the prune operation proposes those too, and the acceptance test decides. On life 11 the removal of 55 atoms was accepted, and of the model and extension proposed beside it, the extension was rejected on an interval wholly below zero and the model on one that included zero. The language went from 426 to 371 atoms, and untrained combinations recovered to 18.6 against a start of 20.5, with every other pool and area above its start.

Two further revisions went in at the same version. First, the acceptance rule had been "solve strictly more tasks than the current state", and on life 10 a model was accepted for a gain of 24 tasks on 72 worlds that read 82 *lower* on the untouched set; the bootstrap interval replaced the strict inequality. Second, the tie-break, the order the scan uses among rows it scores equally, had been a random shuffle of row positions, so adding or removing any atom reshuffled every other row and every language test carried that noise. It is now a fingerprint of each row's content, so a language change changes nothing but the rows it adds or removes.

8.3 A gate missed at its own noise level

Seed 919116 under version 9 failed the growth gate in `drawn7` at +8% against the required 15%. The question was whether something had gone wrong or whether the check was too noisy to tell. Replaying that area's fresh 24-world set with the state after each accepted change answered it: the score stood at 258 at the start, 283 after the area's own accepted change (+10%), and between 259 and 283 under every later shared-language change, ending at 279. Not cumulative drift: one accepted change whose gain on its 48-world test set (+35%) generalised to +10% on the fresh set, after which the area's later candidates, at +7% to +10% on 48 worlds, were rejected because their intervals included zero.

The acceptance test can detect a 15% gain in a single-pool area on 48 worlds only barely. Version 10's extension set is the response: a candidate that lands near the threshold gets a second sample of the same size before it is rejected, and the decision uses both.

8.4 Smaller things

The test cost six times the living. Version 1's remedy was structural: longer lives, held-out sets reused until something is accepted on them, and stopping a losing candidate at the half set.

Two bookkeeping flaws. Two parts of one life accepted on the same set (version 2), and a cached baseline that went stale as experience grew (version 3). Neither changed a decision. Both were caught only because the audit replays everything.

A guard that could not do its job. Version 5's per-candidate tolerance on unguided search let three candidates, each inside tolerance, add up to an 11% dip. A later check across all nineteen seeds found no cumulative drift in the pool the guard was meant to protect, so no replacement guard was added.

A filter that excluded the right answers. The string worker's consistency test rejected pieces that the puzzle generator itself uses. A tenth of the tasks were unsolvable in principle until version 9.

The host machine. One stall, one crash and one interrupted run in the record were the machine's memory, not the agent.

9 What this does and does not show

It shows one program improving its own state in seven settings of different structure, across two kinds of puzzle, keeping only changes that it tested on worlds it never trained on, with every test charged and every acceptance independently replayed. The three forms of growth from the earlier work appear together in its ledger: within-life accumulation in every life, learned search guidance accepted nearly every life, and invented atoms that let it reach families its starting language could not express. The arms of Section 7.5 measure the last two: without guidance the agent kept 6% of its growth, and without invention 65%. Growth in one area did not require starting over in another; the language carried from area to area.

It does not show that testing changes is what makes the agent grow. At this budget and in these seven areas, keeping every proposed change grew as much for less than a third of the computing. The test's measured value is safety: it kept far fewer changes that made things worse. Whether that balance shifts with more areas, a smaller budget or longer runs is not known.

It is improvement of state, not of procedure. The agent's model and language direct its search, its search produces the experience the next model and language are fitted from, and the ledger shows the state improving the state across many lives. But the procedures that turn experience into a model, candidates into atoms, and tests into acceptances are fixed code, and every revision of the acceptance rule was made by the experimenter. A later agent does not produce a better reflect or a better extend. That was true through version 10. In version 11.1 the agent does change part of how it learns: it turns its own dials, from evidence in its own ledger, under a test nothing else used, and it lives under what it chose, with every choice replayed. That is the first rung of improving the improver. It is a low rung. The dials and their settings were written by the experimenter; the agent picks among them and cannot invent a new rule. Across the ten seeds of the final version, twelve of its thirteen adoptions made learning cheaper, not better, and the one that made it better rests on a single test that cleared zero by a small margin. And the part that turns the dials is itself fixed. So the programme's definition of recursive self-improvement is met only in this narrow sense.

The growth also saturates and is uneven. On the main agent's later mechanism lives few candidates passed, and model gains fell from hundreds of tasks to single digits by the sixth life. Half of the remaining invented atoms are behaviourally new and some are surely junk that costs budget until a removal takes them out. Testing costs two to three times living. The two puzzle forms are both small and synthetic, and no setting involves a real-world domain. A reviewer suggested testing the same idea on a coding assistant built on a large language model. That would be a different system, because this agent has no language model and none of its parts was designed for one, so it is left outside this paper.

The held-out sets behind the tables are single draws whose per-world results were not saved, so no interval accompanies the totals; the figure's points are totals only. Two accepted changes of a development agent's run do not replay, for a reason not found.

9.1 Limits a reader should know

Before this paper was posted, it was audited against its own repository. The audit's list of ways the results could mislead is given in full in the technical companion; in plain terms it comes to this.

- **The starting point is held back on purpose.** The agent starts without two of the nine families of building blocks, and every area uses them, so part of its measured growth is getting back what it was denied. A starting point given every building block, and nothing learned, already reaches about a third of the full agent's growth (30% to 44% on the three seeds measured), most of it in the two metric areas. The full agent ends well ahead of that starting point on all three seeds. The growth percentages in this paper are measured from the held-back start.
- **Each score is one sample.** Every score in the tables comes from one sample of puzzles. The arms were scored a second time on a fresh sample, with the script committed before it ran. Every conclusion about the arms came out the same: guidance and invention matter, the test adds no growth, and spending the test's computing on experience grows more. But single areas moved a great deal between the two samples (one from +42% to +134%), and one group of puzzles that rose on the first sample fell to 91% of its start on the second. "Every group rose on every seed" is true of the first sample only.
- **Ten seeds, but the same seven areas.** Every seed meets the same three drawn areas and the same two metric areas. A seed changes which worlds are drawn, not how an area is built, so the results say nothing about a new area.
- **The metric areas are small.** The two metric areas contain only 118 and 80 different worlds, and they show the largest gains in this paper. An agent that lives a few hundred worlds meets most of them, so 77% to 98% of the metric worlds it was scored on were worlds it had already lived. The matched arm lived every one. The gains on the few worlds the agent had never lived were as large as on the rest.
- **"Harmful" is a rough label.** A kept change is counted as harmful if its area's score fell on one small sample. Many such falls are within the noise of that sample, and three seeds are too few to claim that 2 in 35 is reliably different from 15 in 63.
- **Equal computing is not equal time.** The matched arm used the same number of scans as the full agent and two and a half times its running time, because scans do not count everything the program does.
- **The agent's changes to its own dials are a small result.** Twelve of the thirteen came from tests in which the changed dial made no difference to any puzzle; they were kept because the tested round cost less, and over a whole run they saved nothing. The thirteenth is one pass among ten tests in which the dial did matter, about what chance alone would sometimes produce.
- **One kind of computer.** The runs replay exactly on the machine type that made them. On Linux, with the same software versions, two of the seven areas draw different worlds, so a rerun there is a fresh experiment, not a copy.

- **The rules were revised along the way.** Seven of the eleven revisions of the procedure followed a failed condition and four did not, and one threshold was moved after its results had been read; the runs it was moved for are counted as failures.

10 How this relates to other work

The technical companion paper gives the references; this is the short version. Research on *continual learning* asks how a program can keep what it learned while learning new things, usually by restricting how its training changes it. Here a change is instead tried everywhere it reaches and refused if it does harm. Research on *program synthesis* has long used learned guidance to decide what to try first, and the closest earlier system, DreamCoder, grows a library of building blocks from programs it has solved. The agent here invents building blocks from puzzles it could not solve, and each must pass a held-out test before it is kept. Other recent systems, such as the Darwin Gödel Machine, change far more of their own code than this agent does. What this paper adds is on the side of evidence: conditions stated in advance, sealed seeds, charged test costs and a replay of every decision.

11 Lessons

1. Guidance and invention are the engine; the test is the brake. Without either of the first two the growth fell on every seed. Keeping every proposed change grew as much as testing them, for less than a third of the computing; what the test bought was 2 harmful changes kept out of 35 instead of 15 out of 63. Measure what a test is for before calling it the design.

2. Test the combination first, then the parts. This halved the tests in the common case; whether it loses anything against testing every part was not measured.

3. Invention needs recurrence. An atom that fits one uncovered task is usually a coincidence; one that fits two is usually a missing atom.

4. Let the agent remove what it added. Removals of atoms that never earned their keep were accepted three times in four, and removing the atoms the self model called import-only was accepted with a gain in all five areas. It did not by itself undo the loss on untrained combinations; most of that apparent recovery was a change of baseline.

5. Shared state needs shared tests. The 10% loss in an untouched area was invisible to any single-area test.

6. Score growth on worlds the agent never saw, and attach an interval. A gain of 24 on 72 worlds is noise.

7. Make the measurement stable under the change being measured. A tie-break over row positions turned every language change into a reshuffle of the whole catalog.

8. A gate at the noise level of its check will fail by chance. Give borderline candidates a second sample before rejecting them.

9. Freeze, seal, forecast, replay. The audit found both bookkeeping flaws of the confirmation phase, and neither would have been found by looking at the scores.

12 What comes next

- The arms on more seeds, with intervals, and at other budgets and numbers of areas, to find where, if anywhere, the test pays for itself in growth and not only in safety; and a cheaper check aimed at harm alone.
- An eighth setting: a second string-task area with a different split of trained and untrained key families, already smoke-tested with no new code, as version 12 on two new seeds.
- Intervals on the reported gains, by saving per-world results in the final checks.
- Past the fixed dials: letting the agent propose new pieces of its own rules, such as a new way to vote or how large a test each kind of change needs, under the same dial test; and repeating the better-not-cheaper adoption, which so far rests on one test.

13 Glossary

Acceptance test ("prove")
: The comparison that decides whether a proposed change is kept: both states run on the same held-out worlds, and the change is accepted only if the 95% bootstrap interval of its per-world gain lies above zero and, for a language change, it stays within 2% on every other lived area of the same form.

Area
: A recipe for generating worlds with a particular hidden structure. Seven are used: mechanism worlds, three drawn partitions, two metric weightings and string tasks.

Atom
: One simple yes-or-no rule about a character, the smallest building block. 313 exist in 9 families; the agent starts with 260.

Audit
: An independent replay of every accepted change from saved state, checking that the recorded totals and interval come out identical.

Bootstrap interval
: A range estimated by resampling the observed per-world differences with replacement many times (4,000 here) and taking the middle 95% of the resampled averages. If it lies entirely above zero, the gain is unlikely to be luck.

Carried state
: Everything the agent keeps between lives: language, experience, per-area model and prior, invented atoms, ledger and resource account.

Catalog
: Every candidate rule the current language can build, one row per distinct behaviour. 82,758 rows from the starting language.

Confident pick
: A submission made when exactly one consistent rule lay in the scanned window. Right about 90% of the time; not a guarantee.

Cross-area check
: The part of the acceptance test that runs a language change on a held-out set of every other lived area of the same form and rejects it if it falls more than 2% behind there.

Discovery rate
: How often the agent finds an atom when a world contains it, as estimated by its own model. An atom below 0.10 is filed in no group.

Earned experience
: The atoms behind the agent's own submissions in a world, kept whether the submission was right or wrong.

Extension set
: A second held-out set of the same size, drawn fresh, on which a candidate near the acceptance threshold is run before the decision is made on both sets together. Added in version 10.

Family
: One of 9 kinds of atom, named for its arithmetic operation. Two families, popcount and xormod, are withheld from the starting language.

Gate
: A pass-or-fail condition stated before a confirmation run. The four are growth (every area up at least 15%), safety (no pool below 90% of its start), cost (testing at most a stated multiple of living) and audit (every acceptance replays exactly, on its own set).

Hash
: A short fingerprint computed from a file's exact contents. The frozen protocol records the hashes of the source files.

Held-out set
: Worlds the agent never lived, used only to compare two states. Fresh held-out sets, drawn with a salt no test used, give the reported gains.

Import rate
: How often the agent earns an atom through a wrong pick in a world that does not contain it, as estimated by its own model.

Language
: The set of atoms the agent currently holds. Not a natural language.

Life
: One round of experience (48 or 96 worlds of one area, 100 tasks each) followed by proposing and testing changes.

Paired comparison
: Comparing two states on the same worlds and recording the per-world difference, so that variation in world difficulty cancels.

Pool
: A subset of an area's worlds that the scoring can tell apart but the agent cannot: trained, untrained and unfamiliar in mechanism worlds and string tasks.

Prior (search prior)
A starting order for the catalog based on how often atoms appeared together in the agent's own earlier worlds of an area. Half of each task's budget is reserved for the order the agent would use without it.

Random seed
A number that fixes every pseudorandom choice a run makes, so the run can be repeated exactly. Sealed seeds were chosen in advance and never used before.

Replay
Rebuilding the state before and after an accepted change and running the same test again. "Replays exactly" means the totals and interval match the ledger.

Salt An extra number mixed into a random draw so that one held-out set differs from every other.

Scan
Looking at one candidate rule. The budget is 1,024 scans per task.

Seed agent
The program plus its carried state. The thing that grows.

State and procedure
State is what the program carries and changes; procedure is the fixed code that reads and writes it. This work improves state only.

Task
One puzzle: a hidden rule, eight visible examples (three in the string form) and sixteen hidden ones (six in the string form).

Tie-break
The order the scan uses among catalog rows it scores equally. Now a fingerprint of each row's content, so that a language change does not reshuffle unrelated rows.

World
A hidden set of 14 atoms and 100 tasks whose rules are built only from them.

World and self model
An area's fitted model of which atoms occur together (world) and how reliably the agent finds each (self). It guides search and supplies the import-only verdict that prune uses.

14 Reproducibility

The technical companion to this paper, with the same results in the field's own terms, is `paper/seed-growth.tex`. The repository is `https://github.com/MMWilliams/rsi`, and this paper corresponds to its tag `seed-growth-arxiv-v1`. The code is released under the MIT licence and this paper under CC BY 4.0. The companion gives the reference platform, the environment, and what the records do and do not cover. One command, `shreproduce_tables.sh`, rebuilds every generated table from the committed runs without running the agent and fails if anything differs. The seed agent is `rsi/seed.py` (state, areas, the operations and the acceptance test) and `rsi/seed_string.py` (the string-task form), with `rsi/seed_areas.py` for metric worlds, `rsi/invention.py` for the grammar and the recurrence test, and `rsi/self_world_model.py` for the model. `seed_run.py` runs lives and

resumes from saved state; `seed_check.py` scores an agent on fresh held-out sets; `audit_seed.py` replays accepted changes from version snapshots; `confirm_seed.sh<name><seed>` runs the frozen confirmation schedule, check and audit described in `EXPERIMENT-seed-confirmation.md`, whose runs are in `seeds/c1` to `seeds/c31` (there is no `seeds/c20`: that seed was interrupted, never rerun and left no data); `summarize_v11_replication.py`, `rejudge_proofs.py`, `summarize_arms.py` and `paper_tables.py` produce the ten-seed, re-testing and arms tables; the arms themselves are `rsi/seed_arms.py` and `run_arms.py`, and their rules were committed in the commit tagged `prereg-review002-arms`. States are in `seeds/<name>/state.json`; the ledger inside each is the full record of every candidate, its test and its cost. The worked task of Section 2 is reproduced by drawing the first trained world of `MechanismArea(644101)` with `random.Random(7)` and the task from its first mechanism with `random.Random(11)`.

Settings: 48 worlds of 100 tasks per life in development and 96 in confirmation; budget 1,024 scans per task; held-out sets of 24 worlds per pool in development and at least 48 in confirmation, cross-area sets of 12 and 24; tolerance 2% relative; 6 to 8 groups and 2 to 4 groups per world by held-out likelihood; extension cap 40 atoms per life with recurrence 2; removal after 2 lives or on the self model's import-only verdict; tie-break a content hash per row with seed 77; a change accepted only when the paired 95% bootstrap interval (4,000 draws) of its gain is above zero, or, from version 10, when the interval over the proof set and one extension set is.

Use of AI tools. An AI coding assistant (Claude, by Anthropic) was used to write the code and to draft this text under the author's direction, and the author is responsible for the content.

Table 3: The twelve confirmation versions. "Areas" is the number of areas lived and checked. The cost figure is testing as a multiple of living.

Version	Seeds	Areas	What changed from the version before	Outcome on the two seeds
1	919100, 919101	3	First frozen protocol.	Growth, safety and audit pass; cost fails at 5.96 and 5.88.
2	919102, 919103	3	Lives of 96 worlds, held-out sets of 48, a set kept until something is accepted on it, stop at the half set when the gain is not positive.	Cost 1.89 and 1.98; one seed fails the clause that each acceptance has its own set.
3	919104, 919105	3	A fresh set after any acceptance within a life.	One stale recorded baseline per seed (47 and 27 tasks); every acceptance stands.
4	919106, 919107	3	Cached baseline keyed on the experience count too.	All four gates pass, at 2.32 and 2.04. Confirmed for three settings.
5	919108, 919109	3	A guard: a language candidate is also scored with model and prior off and rejected if that unguided score falls more than 10%.	Growth and safety fail on one seed, cost fails on the other (2.53). Retired.
6	919110, 919111	6	Version 4's code, three settings new to every agent.	All twelve area scores and twenty pools rise; cost fails at 2.78 and 2.64 against 2.5.
7	919112, 919113	6	Cost gate 2.5 plus 0.25 per setting beyond three (3.25 at six).	All four gates pass, at 2.88 and 2.32.
8	919114, 919115	7	The string-task area, a second puzzle form; gate 3.5.	All four gates pass, at 2.61 and 2.71.
9	919116, 919117	7	The string worker's consistency filter repaired.	One seed fails growth in `drawn7` at +8%; the other passes all four.
10	919118, 919119	7	Extension set for a candidate at the acceptance threshold.	First seed passes all four at 2.92; second interrupted by the host machine at life 20 of 21, clean rerun pending.
11	919120	7	The agent tests changes to six dials of its own rules (Section 7.3); a fifth gate, at least one dial change adopted.	Growth, safety, cost and the fifth gate pass; replay fails on one dial test whose losing copy's experience was not saved.
11.1	919121, 919122	7	That experience is saved; nothing the agent decides changes.	All five gates pass on both, at 2.56 and 3.67; four of eight dial changes adopted, one for a better state.

Table 4: Three-setting confirmation runs: tasks solved on fresh held-out sets (24 worlds per pool; 7,200, 2,400 and 4,800 tasks) by the starting and the retained state, the test-to-live cost ratio, and the gate outcome.

Seed (version)	mech	drawn7	metric	cost	gates
919100 (v1)	1,128 to 1,997	236 to 385	405 to 1,055	5.96×	cost fails
919101 (v1)	1,223 to 1,944	271 to 318	318 to 1,619	5.88×	cost fails
919102 (v2)	1,020 to 2,010	281 to 409	364 to 1,550	1.89×	own-set clause fails
919103 (v2)	1,200 to 1,697	255 to 363	363 to 1,428	1.98×	all pass
919104 (v3)	1,276 to 1,807	280 to 364	373 to 1,927	2.29×	one stale baseline
919105 (v3)	1,147 to 2,366	180 to 342	322 to 1,518	1.94×	one stale baseline
919106 (v4)	1,129 to 2,136	202 to 299	343 to 1,647	2.32×	all pass
919107 (v4)	1,138 to 2,263	275 to 382	248 to 1,901	2.04×	all pass
919108 (v5)	1,179 to 1,990	367 to 388	281 to 1,872	2.48×	growth and safety fail
919109 (v5)	1,111 to 2,151	278 to 334	404 to 1,626	2.53×	cost fails

Table 5: Six- and seven-setting confirmation runs (versions 6 to 10): tasks solved on fresh held-out sets (24 worlds per pool) by the starting and the retained state in each area, the test-to-live cost ratio, and the gate outcome. The cost gate was 2.5 for version 6, 3.25 for version 7 and 3.5 for versions 8 to 10.

Seed (version)	mech	drawn7	metric777001	drawn11	drawn13	metric555001	str7	cost, gates
919110 (v6)	1,197 to 2,442	305 to 375	299 to 1,415	291 to 424	375 to 483	356 to 1,833	–	2.78×, cost fails
919111 (v6)	1,187 to 2,160	305 to 384	355 to 1,585	206 to 341	176 to 314	283 to 1,475	–	2.64×, cost fails
919112 (v7)	1,014 to 2,070	214 to 374	493 to 1,723	217 to 415	237 to 375	308 to 1,390	–	2.88×, all pass
919113 (v7)	1,269 to 1,878	230 to 334	389 to 1,667	249 to 369	258 to 375	284 to 1,823	–	2.32×, all pass
919114 (v8)	1,185 to 2,300	166 to 460	344 to 1,422	249 to 381	215 to 439	371 to 1,605	4,749 to 6,026	2.61×, all pass
919115 (v8)	1,174 to 1,885	320 to 413	309 to 1,409	251 to 386	322 to 430	300 to 1,321	4,119 to 5,419	2.71×, all pass
919116 (v9)	1,072 to 2,088	258 to 279	259 to 1,715	220 to 377	224 to 384	339 to 1,485	5,076 to 6,248	2.33×, growth fails
919117 (v9)	1,217 to 2,416	241 to 432	326 to 1,584	275 to 416	305 to 442	384 to 1,716	4,910 to 6,127	2.27×, all pass
919118 (v10)	1,159 to 2,484	273 to 349	450 to 2,283	241 to 439	286 to 379	378 to 1,329	4,722 to 6,193	2.92×, all pass
919120 (v11)	1,164 to 2,185	362 to 463	364 to 1,522	251 to 368	215 to 338	410 to 1,253	4,871 to 6,278	4.12×, replay fails
919121 (v11.1)	1,309 to 2,356	151 to 326	485 to 1,497	225 to 413	261 to 434	360 to 1,454	4,386 to 6,014	2.56×, all pass
919122 (v11.1)	1,166 to 2,385	260 to 410	259 to 1,893	303 to 454	214 to 410	458 to 1,764	4,569 to 5,873	3.67×, all pass

Table 6: Version 11.1 on ten sealed seeds: change from the starting state to the retained state on fresh held-out sets, by area and pool. Median and minimum over seeds, 95% bootstrap interval of the median (1,000,000 resamples of seeds), and the number of seeds below the start. Each seed contributes one draw of 24 worlds per pool; the interval is over random seeds for the fixed area structures.

Area or pool	Median	Weakest seed	95% interval	Seeds below start
mechanism worlds	+84%	+64%	+77% to +90%	0
trained	+51%	+22%	+43% to +58%	0
unfamiliar	+410%	+165%	+347% to +585%	0
untrained	+67%	+13%	+45% to +91%	0
drawn7	+61%	+21%	+49% to +91%	0
metric777001	+431%	+209%	+264% to +511%	0
seen centre	+311%	+147%	+198% to +358%	0
unfamiliar centre	+802%	+406%	+478% to +1075%	0
drawn11	+48%	+18%	+24% to +75%	0
drawn13	+66%	+5%	+51% to +92%	0
metric555001	+306%	+137%	+269% to +359%	0
seen centre	+222%	+89%	+187% to +268%	0
unfamiliar centre	+652%	+394%	+507% to +1678%	0
str7	+32%	+20%	+29% to +35%	0
trained	+21%	+10%	+16% to +26%	0
unfamiliar	+66%	+49%	+55% to +69%	0
untrained	+19%	+8%	+15% to +23%	0

Table 7: Version 11.1 per seed: cost of level-0 proofs and of all proofs relative to living (limits 3.5 and 5.0), accepted changes, all of which replayed, procedure changes adopted, and the gate outcome.

Seed	Level-0	All proofs	Accepted	Adopted	Gates
919121	2.10×	2.56×	11	1	all pass
919122	2.71×	3.67×	14	3	all pass
919127	2.61×	3.63×	12	0	growth fails
919128	2.30×	3.05×	10	1	all pass
919129	2.97×	4.04×	13	1	all pass
919130	2.35×	3.43×	11	2	all pass
919131	2.76×	3.52×	16	1	all pass
919132	2.26×	3.02×	10	1	all pass
919133	3.04×	4.15×	13	2	all pass
919134	2.51×	3.68×	13	1	all pass

Table 8: Recorded full-set proofs of the sealed seeds judged again on part of their own set: full-set interval passes the smaller set would lose (a pass is kept only if it also clears the cross-area check) and full-set interval failures it would pass.

Set size	Passes lost	Failures passed
half	125 of 435 (29%)	30 of 467 (6%)
quarter	207 of 435 (48%)	37 of 467 (8%)

Table 9: The review-002 arms on seeds 919127 to 919129, meta-level off. G is the mean over the seven areas of ln(retained/start) on fresh held-out sets; the difference from full is per seed; scans are live plus proof, mean over seeds, in millions; harmful counts kept changes that lowered their own area on a fresh set. Outcomes follow the rules committed before any arm ran; invention is judged on the unfamiliar-family pools of mech and str7, not on G. Brier score of the eight committed forecasts 0.126.

Arm	G	G − full, per seed	Scans	Kept	Harmful	Outcome
full	0.695	–	717	12, 10, 13	2 of 35	–
accept-all	0.691	+0.045, −0.027, −0.030	206	21, 21, 21	15 of 63	rule 3: inconclusive
point	0.706	+0.066, −0.023, −0.009	366	18, 21, 18	17 of 57	rule 3: inconclusive
matched	0.791	+0.150, +0.059, +0.080	721	76, 70, 74	–	rule 2: withdraw
no-guidance	0.045	−0.650, −0.625, −0.676	341	0, 1, 0	–	rule 4: shown
frequency	0.513	−0.128, −0.184, −0.235	666	9, 10, 6	–	rule 4: shown
no-invention	0.450	−0.181, −0.290, −0.265	431	11, 9, 9	–	rule 4: shown

Table 10: The arms scored again on a second evaluation draw (salt 20001, 24 worlds per pool) that nothing had used. G as in Table 9, mean over the three seeds, on the first and the second draw; then the difference from full on the second draw, per seed. A star marks a difference whose paired 95% bootstrap interval over worlds (10,000 resamples within pools) excludes zero. The last row is the starting state given every generator atom, with no model and no prior.

Arm	G, first draw	G, second draw	G − full on the second draw, per seed
full	0.695	0.763	–
accept-all	0.691	0.771	−0.003, +0.018, +0.006
point	0.706	0.771	+0.002, +0.023, −0.003
matched	0.791	0.879	+0.139*, +0.125*, +0.082*
no-guidance	0.045	0.057	−0.737*, −0.561*, −0.821*
no-invention	0.450	0.469	−0.280*, −0.300*, −0.304*
frequency	0.513	0.569	−0.116*, −0.192*, −0.275*
start, nothing withheld	–	0.271	–

Table 11: Ablation of `s1`'s retained state (version 24) on the mechanism area's untouched held-out set: tasks per world by pool, and total of 7,200.

State	Trained	Unfamiliar	Untrained	Total
retained	49.5	18.7	18.2	2,072
retained, no prior	43.0	14.7	9.5	1,614
retained, no model	37.3	15.0	15.9	1,639
grown language only, no model, no prior	11.1	9.0	10.5	735
starting language, retained model and prior	50.0	6.1	26.2	1,973
starting state	26.2	3.7	24.2	1,299